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Massager

Abstract

A massager includes a massager body, a vibrating assembly, and an expansion massage structure. The massager body includes a mounting portion and a massage portion. The mounting portion defines a mounting cavity. The massage portion defines an accommodating cavity. The massage portion is configured to insert into a human body cavity to perform massage. The vibrating assembly is mounted in the accommodating cavity and drives the massage portion to vibrate. The expansion massage structure includes a pneumatic system and at least one expansion portion. The pneumatic system is mounted in the mounting cavity and inflates and deflate at least one expansion portion. The at least one expansion portion is disposed on a periphery of the massage portion or one end of the massage portion away from the mounting portion, and the at least one expansion portion is configured to insert into the human body cavity to perform massage.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a technical field of massager devices, and in particular to a massager.

BACKGROUND

(2) Currently, most massagers on the market use a single direct-current (DC) motor to vibrate to realize a massage function. Such massagers have a relatively simple function, lacks diversity and adaptability, and easily leads to a poor user experience.

SUMMARY

(3) Embodiments of the present disclosure provide a massager, which uses a massage portion and at least one airbag to massage different positions of a human body cavity, thereby improving a user experience.

(4) The present disclosure provides a massage. The massager comprises a massager body, a vibrating assembly, and an expansion massage structure. The massager body comprises a mounting portion and a massage portion connected with the mounting portion. The mounting portion defines a mounting cavity. The massage portion defines an accommodating cavity. The massage portion is configured to insert into a human body cavity to massage the human body cavity. The vibrating assembly is mounted in the accommodating cavity and is configured to drive the massage portion to vibrate. The expansion massage structure comprises a pneumatic system and at least one expansion portion. The pneumatic system is mounted in the mounting cavity and is configured to drive the at least one expansion portion to expand and contract. The at least one expansion portion is disposed on one end of the massage portion away from the mounting portion, and the at least one expansion portion is configured to insert into the human body cavity and massage the human body cavity.

(5) In the present disclosure, the massage portion and the at least one expansion portion are configured to insert into the human body cavity and massage the human body cavity. The vibrating assembly drives the massage portion to vibrate, and the pneumatic system drives the at least one expansion portion to expand and contract. During an expansion process of the expansion portion, the human body cavity are compressed, thereby realizing a massage function. The at least one expansion portion is disposed on one side of the massage portion. In this way, the massage portion and the at least one expansion portion massage different positions of the human body cavity, thereby improving the user experience.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) In order to clearly describe technical solutions in the embodiments of the present disclosure, the following will briefly introduce the drawings that need to be used in the description of the embodiments or the prior art. Apparently, the drawings in the following description are merely some of the embodiments of the present disclosure, and those skilled in the art are able to obtain other drawings according to the drawings without contributing any inventive labor.

(2) FIG. **1** is a schematic diagram of a massager according to a first embodiment of the present disclosure.

(3) FIG. **2** is a schematic diagram of the massager according to a second embodiment of the present

disclosure.

(4) FIG. **3** is a schematic diagram of the massager according to a third embodiment of the present disclosure.

(5) FIG. **4** is a schematic diagram of the massager according to a fourth embodiment of the present disclosure.

(6) FIG. **5** is a schematic diagram of the massager according to a fifth embodiment of the present disclosure.

(7) FIG. **6** is a cross-sectional schematic diagram of the massager of the present disclosure where an extension portion thereof is shown in a first alternative configuration.

(8) FIG. **7** is a cross-sectional schematic diagram of the massager of the present disclosure where the extension portion is shown in a second alternative configuration.

(9) FIG. **8** is a cross-sectional schematic diagram of the massager of the present disclosure where the extension portion is shown in a third alternative configuration.

(10) FIG. **9** is a cross-sectional schematic diagram of the massager of the present disclosure where the extension portion is shown in a fourth alternative configuration.

(11) FIG. **10**A is an exploded schematic diagram of a massager body of the massager of the present disclosure; and FIG. **10**B is a schematic view of a flexible sleeve that sleeves the massager body of the present disclosure.

(12) FIG. **11**A is a cross-sectional schematic diagram of the flexible sleeve of the present disclosure; and FIG. **11**B is an exploded cross-sectional schematic view of the massager body of the present disclosure.

(13) FIG. **12** is a cross-sectional schematic diagram of a pressure relief valve of the present disclosure.

(14) FIG. **13** is a schematic diagram of a massager body assembled with a pneumatic system of the present disclosure.

(15) FIG. **14** is an exploded schematic diagram of the massager body and the pneumatic system of the present disclosure.

(16) FIG. **15** is an exploded schematic diagram of a massage portion of the present disclosure.

(17) FIG. **16** is a schematic diagram of the massager according to a sixth embodiment of the present disclosure.

(18) FIG. **17** is a schematic diagram of the massager according to a seventh embodiment of the present disclosure.

(19) FIG. **18** is a schematic diagram of the massager according to an eighth embodiment of the present disclosure.

(20) FIG. **19** is a schematic diagram of the massager according to a ninth embodiment of the present disclosure.

REFERENCE NUMBERS IN THE DRAWINGS

(21) **100**—massager body; **110**—mounting portion; **111**—first housing; **112**—second housing; **113**—mounting cavity; **120**—massage portion; **121**—accommodating cavity; **122**—first surface; **123**—wiring channel; **124**—third portion; **125**—fourth portion; **130**—first snapping structure; **140**—activity space; **200**—vibrating assembly; **300**—expansion massage structure; **310**—pneumatic system; **311**—air pump; **312**—control valve; **313**—connecting tube; **314**—communicating tube; **315**—pressure relief valve; **3151**—valve housing; **3152**—valve core; **3153**—reset piece; **316**—connecting piece; **3161**—first channel; **3162**—second channel; **3163**—third channel; **320**—expansion portion; **321** airbag; **322**—protection portion; **322***a*—mounting space; **400**—flexible sleeve; **410**—main portion; **411**—first assembly cavity; **412**—first portion; **413**—second portion; **430**—vibrating portion; **431**—second assembly cavity; **450**—gap; **460**—second snapping structure; **500**—vibrating structure; **600**—control board; **700**—battery; **800**—button assembly.

DETAILED DESCRIPTION

(22) In order to make objectives, technical solutions, and advantages of the present disclosure

clearer, the following further describes the present disclosure in detail with reference to accompanying drawings and embodiments. It should be understood that the specific embodiments described here are only used to explain the present disclosure, but not to limit the present disclosure.
(23) As shown in FIGS. **1-6**, in one embodiment, the present disclosure provides a massage. The massager comprises a massager body **100**, a vibrating assembly **200**, and an expansion massage structure **300**.
(24) As shown in FIGS. **2** and **3**, the massager body **100** comprises a mounting portion **110** and a massage portion **120** connected with the mounting portion **110**. The mounting portion **110** defines a mounting cavity **113**. The massage portion **120** defines an accommodating cavity **121**. The vibrating assembly **200** is mounted in the accommodating cavity **121** and is configured to drive the massage portion **120** to vibrate. The massage portion **120** is configured to insert into a human body cavity to massage the human body cavity.
(25) As shown in FIG. **6**, in one embodiment, the massage portion **120** is a silicone keel, the silicone keel is of a columnar structure, and the accommodating cavity **121** is defined in an interior of the massage portion **120**. The silicone keel has good toughness and elasticity to shape a specific shape and structure, so as to improve a user experience. Of course, in other embodiments, the massage portion **120** may also be made from rubber. In one embodiment, the accommodating cavity **121** extends in a length direction of the massage portion **120**, and the vibrating assembly **200** is slidable in the accommodating cavity **121**. In one embodiment, a width of the accommodating cavity **121** is slightly less than a size of the vibrating assembly **200**, so that the vibrating assembly **200** is fixed at different positions in the accommodating cavity **121** through a tension of the massage portion **120**. Therefore, the massage portion **120** is able to vibrate and massage different positions of the human body cavity. However, a specific form of the massage portion **120** is not limited thereto.
(26) As shown in FIG. **6**, in one embodiment, the expansion massage structure **300** comprises at least one pneumatic system **310** and at least one expansion portion **320**. The at least one pneumatic system **310** is mounted in the mounting cavity **113**. The at least one expansion portion **320** is disposed on one end of the massage portion **120**. Specifically, the at least one expansion portion **320** is disposed on the one end of the massage portion **120** away from the mounting portion **110**, That is, the at least one expansion portion **320** is disposed on a top portion of the massage portion **120**. Alternatively, the at least one expansion portion **320** comprises two expansion portions **320**, and the two expansion portions **320** are respectively disposed on a periphery of the massage portion **120** and one end of the massage portion **120** away from the mounting portion **110**. However, positions and numbers of the expansion portions **320** are not limited thereto. The at least one expansion portions **320** is configured to insert into the human body cavity and massage the human body cavity.
(27) The at least one expansion portion **320** comprises at least one airbag **321**. During use, the at least one pneumatic system **310** inflates and deflates the at least one airbag **321** to drive the at least one airbag **321** to expand and contract. During an expansion process of the at least one airbag **321**, the at least one airbag **321** extrude the human body cavity, thereby playing a massage role. The at least one pneumatic system **310** repeatedly inflates and deflates the at least one airbag **321** to generate a continuous massage effect. In one embodiment, by controlling a speed and a frequency of inflation and deflation of the at least one airbag **321**, the user is able to adjust a massage intensity and a method of the massager according to personal preferences and physical feelings, thereby obtaining a more personalized massage experience.
(28) It should be noted that, during use, the user is able to hold the mounting portion **110** to adjust a depth of the massage portion **120** and the expansion massage structure **300** inserting into the human body cavity to improve the massage effect.
(29) As shown in FIGS. **1-5**, in some embodiments, the massager body **100** has a front surface, side

surfaces, and a rear surface. The at least one expansion portion **320** is disposed on the front surface of the massager body **100**. Alternatively, the at least one expansion portion **320** is disposed on any one of the side surfaces of the massager body **100**. Alternatively, the at least one expansion portion **320** is disposed on the rear surface of the massager body **100**. It should be noted that the at least one expansion portion **320** of the present disclosure comprises one or more expansion portions **320**. When the expansion portions **320** are provided, the expansion portions **320** are disposed on at least two of the front surface, the side surface, the rear surface, and a top portion of the massager body **100**. Of course, the expansion portions **320** may also be disposed on only on one side of the massager body **100**. In this way, the user is able to obtain different massage experiences through the expansion portions **320** disposed in different positions and with different massage intensities.

(30) It should be noted that each of the expansion portions **320** may correspond to a corresponding pneumatic system **310**, or the expansion portions **320** are corresponding to one pneumatic system **310**.

(31) As shown in FIGS. **2** and **3**, in some embodiments, the at least one expansion portion **320** of the present disclosure comprises two expansion portions **320**. The two expansion portions **320** are respectively disposed on the two side surfaces of the massager body **100**, or, the two expansion portions **320** are respectively disposed on the front surface and the rear surface of the massager body **100**.

(32) For ease of illustration, the present disclosure takes an example that the massager comprises only one massager and one expansion portion for further description. As shown in FIG. **6**, in some embodiments, the massager comprises the pneumatic system **310**. The pneumatic system **310** comprises an air pump **311**, a control valve **312**, and a connecting tube **313**. An air outlet of the air pump **311** is communicated with an air inlet of the control valve **312**. An air outlet of the control valve **312** is communicated with the connecting tube **313**. An air outlet of the connecting tube **313** is communicated with the airbag **321**. Air is introduced into the airbag **321** through the air pump **311**, the control valve **312**, and the connecting tube **313** to inflate the airbag **321**. The control valve **312** is communicated with the airbag **321** and an outside, so as to deflate the airbag **321**. In this way, the airbag **321** is repeatedly expanded and contracted, so as to meet a massage requirement of the airbag **321** on the human body cavity. Further, the present disclosure further include a communicating tube **314**, and the communicating tube **314** is communicated with the air pump **311** and the control valve **312**. Both the connecting tube **313** and the communicating tube **314** are made from a flexible material.

(33) As shown in FIGS. **6** and **7**, the massager further comprises a flexible sleeve **400**. The flexible sleeve **400** comprises a main portion. **410**. The main portion **410** defines a first assembly cavity **411**. The first assembly cavity **411** is configured to accommodate the massager body **100**. The airbag **321** is formed on the flexible sleeve **400**, and the airbag **321** is connected with the main portion **410**. In this way, after the flexible sleeve **400** is sleeved on an outer side of the massager body **100**, the airbag **321** is located on the periphery of the massage portion **120** or the one end of the massage portion **120** away from the mounting portion **110**, and a mounting operation thereof is simple. In addition, the flexible sleeve **400** is soft and elastic, and provides a comfortable tactile sensation. The main portion **410** wraps the massager body **100**, so that a use feeling of directly contacting a hard material is reduced, and comfort and pleasure of use are improved. In addition, the main portion **410** and the airbag **321** are integrally formed on the flexible sleeve **400** by injection molding, which has high processing efficiency. The pneumatic system **310** is communicated with the airbag **321** and is not communicated with the main portion **410**. In this way, an airtightness of the airbag **321** is improved, and it is more conducive to the pneumatic system **310** to inflate and deflate the airbag **321**.

(34) As shown in FIGS. **10-11**, in some embodiments, the main portion **410** defines a first assembly cavity **411**, and the first assembly cavity **411** is configured to accommodate the massager body **100**. As shown in FIG. **6**, along a length direction h of the flexible sleeve, one side of the airbag **321** is

connected with one side of the main portion **410**, and an opening of the airbag **321** is communicated with the first assembly cavity **411**. In this way, a connection between the airbag **321** and the main portion **410** is highly stable. In the present disclosure, the opening of the airbag **321** is communicated with the first assembly cavity **411**, so that the flexible sleeve **400** has good integrity. Further, the flexible sleeve **400** completely wraps the massager body **100**, making a protection effect good. Of course, in other embodiments, the opening of the airbag **321** is not communicated with the first assembly cavity **411**, and the main portion **410** defines an opening, and the connecting tube **313** of the pneumatic system **310** extends from the opening of the main portion **410** to connect and communicated with the airbag **321**.

(35) As shown in FIG. **7**, in some embodiments, along the length direction h of the flexible sleeve **400**, two opposite ends of the airbag **321** are connected with the main portion **410** to define a gap **450** between a middle [portion of a protection portion **322** and a. base portion of the protection portion **322**.

(36) As shown in FIG. **7**, in some embodiments, the airbag **321** comprises a first end and a second end disposed opposite to the first end. The main portion **410** comprises a first portion **412** and a second portion **413** connected with the first portion **412**. The first portion **412** is configured to sleeve on an outer side of the mounting portion **110**, and the second portion is configured to sleeve on an outer side of the massage portion **120**. The first end of the airbag **321** is connected with the first portion **412**, and the airbag **321** is connected with the main portion **400** and is communicated with the first assembly cavity **411** through the first end of the airbag **321**. The second end of the airbag **321** is connected with the second portion **413**. The gap **450** is formed between the airbag **321** and the main portion **410**. It is understood that the pneumatic system **310** is mounted on the mounting portion **110**, the connecting tube **313** of the pneumatic system **310** extends out of the mounting portion **110**, and the first end of the airbag **321** is accordingly connected with the first portion **412**, so that the connecting tube **313** is connected and communicated with the flexible sleeve **400**. The second end of the airbag **321** is connected with the second portion **413**, so that the airbag **321** is sealed at the second end thereof, and the airbag **321** is relatively fixed in a length direction h of the massager. Therefore, the airbag **321** is able to expand in a thickness direction of the massager, which improves comfort of the airbag **321** for massaging the human body cavity.

(37) As shown in FIGS. **8** and **9**, in some embodiments, the airbag **321** is an independent component, the expansion portion **320** further comprises the protection portion **322** assembled on an outer side of the airbag **321**. The protection portion **322** is formed on the flexible sleeve **400**. The protection portion **322** is connected with the main portion **410** and defines a mounting space **322** *a*. The airbag **321** extends into the mounting space **322***a*. The protection portion **322** is configured to expand and contract under driving of the airbag **321**. In the embodiments, the airbag **321** is the independent component, which improves connection stability between the pneumatic system **310** and the airbag **321**. Moreover, the flexible sleeve **400** is assembled on the outer side of the massager body **100** and is sleeved on the outer side of the airbag **321**, so that the airbag **321** is protected. Furthermore, if the airbag **321** is broken, the protection portion **322** reduces an impact of the airbag **321** on the human body cavity and brings higher security to the user.

(38) As shown in FIG. **8**, in some embodiments, along a length direction h of the flexible sleeve, one side of the protection portion **322** is connected with one side of the main portion **410**, and the mounting space **322***a* is communicated with the first assembly cavity **411**. In this way, the flexible sleeve **400** has better integrity, and the flexible sleeve **400** completely wraps the massager body **100**, so as to achieve better protection effect.

(39) As shown in FIG. **9**, in some embodiments, along the length direction h of the flexible sleeve **400**, two opposite ends of the protection portion **322** are connected with the main portion **410** to define the gap **450** between a middle portion of the protection portion and the main portion. In this way, the expansion portion **320** has a larger deformation to meet massage needs of different users.

(40) It should be noted that, in some embodiments, the airbag **321** disposed in the flexible sleeve

400 and the protection portion **322** of the flexible sleeve **400** are an integrated component. That is, when the integrated component is connected and communicated with the pneumatic system **310**, the integrated component plays a role of the airbag **321** and is inflated or deflated by the pneumatic system **310**, so that the integrated component is able to expand and contract to achieve the massage effect. When the integrated component is not connected with the pneumatic system **310**, the airbag **321** of the integrated component is configured to drive the integrated component to expand and contract. The integrated component is bale to expand and contract to achieve the massage effect while protecting the airbag **321** thereof. The flexible sleeve **400** is made from materials such as silicone or rubber, which is not limited thereto.
(41) As shown in FIGS. **6-9**, in some embodiments, the flexible sleeve **400** further comprises a vibrating portion **430**. The vibrating portion **430** is connected with the main portion **410** and disposed on one side of the main portion **410**. The vibrating portion **430** defines a second assembly cavity **431**. The massager further comprises a vibrating structure **500**. The vibrating structure **500** is mounted in the second assembly cavity **431** and is configured to drive the vibrating portion **430** to vibrate The vibrating portion **430** is configured to perform vibration massage on a specific position of a human body.
(42) The vibrating portion **430** vibrates when driven by the vibrating structure **500**, which brings a better massage experience to the user. In one embodiment, the vibrating portion **430** is disposed on the front surface of the massager body **100**. Of course, in other embodiments, the vibrating portion **430** is disposed on the rear surface of the massager body **100**.
(43) In one embodiment, in order to improve the massage effect of the vibrating portion **430**, one or more protrusions are disposed on one side of the vibrating portion **430** facing the main portion **410**. The present disclosure does not limit the number of the one or more protrusions. In addition, different massage methods are realized by configuring the one or more protrusions into different shapes and structures. For instance, if the one or more protrusions are uneven, a stimulating acupressure massage is provided. If the one or more protrusions are soft balls, a soothing and gentle rolling massage is provided. The one or more protrusions of the present disclosure bring a more concentrated and stimulating massage feeling to the user, and the one or more protrusions deepen the massage effect by stimulating specific acupuncture points or muscle groups.
(44) In one embodiment, the vibrating assembly **200** may comprise a motor, a vibrator, etc. The motor is configured to convert electrical energy into mechanical energy to generate vibration. The vibrator is connected with an output shaft of the motor. The vibrator generates vibration through a rotation of the motor. A structure of the vibrating structure **500** may refer to a structure of the vibrating assembly **200**.
(45) As shown in FIG. **11** and FIG. **13**, in one embodiment, in order to improve mounting stability of the flexible sleeve **400** and the massager body **100**, first snapping structures **130** are disposed on an outer wall of the mounting portion **110**, second snapping structures **460** are disposed on an inner surface of the flexible sleeve **400**, and the first snapping structures **130** are respectively snapped with the second snapping structures **460**. The first snapping structures **130** are respectively snapped with the second snapping structures **460** to effectively prevent the flexible sleeve **400** from shifting due to vibration or other external forces during use. The flexible sleeve **400** is fixed to the massager body **100** by the first snapping structures **130** and the second snapping structures **460**, which simplify a mounting process, improve assembly efficiency, and make the disassembly and assembly of the flexible sleeve **400** simple and quick, thereby reducing maintenance time and cost. Optionally, the first snapping structures **130** are hooks, and the second snapping structures **460** are grooves. Alternatively, the first snapping structures **130** are the grooves, and the second snapping structures **460** are the hooks.
(46) As shown in FIG. **13**, in one embodiment, the massage portion **120** has a first surface **122** of the massage portion **120** bent in an extension direction of the massage portion **120**, so as to form an activity space **140** on one side of the massage portion **120**. The expansion portion **320** is at least

partially located in the activity space **140**.
(47) In this way, the expansion portion **320** is within the activity space **140** in a deflated state. At this time, only the massage portion **120** is used for massage. When the expansion portion **320** is in an inflated state, since the first surface **122** of the massage portion **120** is an arc surface, the first surface **122** of the massage portion **120** plays a certain support and limiting effect on the expansion portion **320**, thereby controlling an expansion direction of the expansion portion **320** and improving the comfort of the user when the expansion massage structure **300** massages the human body.
(48) As shown in FIGS. **11-12**, in one embodiment, in order to improve safety of the airbag **321**, the pneumatic system **310** further comprises a pressure relief valve **315**. The pressure relief valve **315** is communicated with the airbag **321**. When an air pressure in the airbag **321** is greater than a predetermined value, the pressure relief valve **315** is configured to reduce the air pressure in the airbag **321**. The pressure relief valve **315** ensures that the air pressure in the airbag **321** is always within a safe range. When the air pressure in the airbag **321** exceeds the predetermined value, the pressure relief valve **315** automatically opens to release excess air to prevent the airbag **321** from exploding due to overpressure or causing other safety problems.
(49) As shown in FIG. **12**, in one embodiment, the pressure relief valve **315** comprises a valve housing **3151**, a reset piece **3153**, and a valve core **3152**. The valve housing **3151** defines a valve cavity, a first opening, and a second opening. The first opening and the second opening are communicated with the valve cavity. The first opening is communicated with the airbag **321**, and the second opening is communicated with the outside. The reset piece **3153** and the valve core **3152** are mounted in the valve cavity. When driven by the reset piece **3153**, the valve core **3152** is configured to close the first opening. When the air pressure in the airbag **321** is greater than the predetermined value, the valve core **3152** is opened by the air in the airbag **321**. A structure of the pressure relief valve **315** is relatively simple, so manufacturing cost is low and the pressure relief valve **315** is easy to maintain and install. The reset piece **3153** may be a spring, and the valve core **3152** is closed when driven by the spring. Moreover, as long as the air pressure in the airbag **321** is greater than an elastic force of the spring, the spring is compressed, thereby opening the valve core **3152**, and the air is realized outward. The reset piece **3153** of the present disclosure quickly drives the valve core **3152**, so that the pressure relief valve **315** responds quickly and releases the air quickly when the air pressure in the airbag **321** exceeds the predetermined value. Of course, in other embodiments, the reset piece **3153** may be a magnetic piece. The magnetic piece is disposed on one side of the valve cavity, and the valve core **3152** and the magnetic piece are attracted or repelled by a magnetic force. The valve core **3152** close the first opening of the valve cavity under driving of the magnetic force. When the air pressure in the airbag **321** is greater than the magnetic force, the air in the airbag **321** opens the valve core **3152**.
(50) As shown in FIGS. **6-9** and **12**, in one embodiment, in order to facilitate the communication between the control valve **312**, the airbag **321**, and the connecting tube **313**, the pneumatic system **310** further comprises a connecting piece **316**. The connecting piece **316** defines a first channel **3161**, a second channel **3162**, and a third channel **3163**. The first channel **3161**, the second channel **3162**, and the third channel **3163** are communicated with each other. The first channel **3161** is communicated with the connecting tube **313**, the second channel **3162** is communicated with the airbag **321**, and the third channel **3163** is communicated with the pressure relief valve **315**. In this way, the air pump **311** drives the air to inflate the airbag **321** through the communicating tube **314**, the control valve **312**, the connecting tube **313**, the first channel **3161**, and the second channel **3162**. Similarly, the air in the airbag **321** is allowed to be released to the outside through the second channel **3162**, the connecting tube **313** and the control valve **312**. Further, the air in the airbag **321** is also allowed to be released to the outside through the second channel **3162**, the third channel **3163**, and the pressure relief valve **315**.
(51) As shown in FIGS. **13-14**, in one embodiment, the massager further comprises a battery **700**

and a control board **600**. The battery **700** and the control board **600** are mounted in the mounting cavity **113**. The battery **700** is configured to provide power to the pneumatic system **310**. The control board **600** is configured to adjust a working condition of the pneumatic system **310**. In one embodiment, the control board **600** is configured to adjust an output pressure of the pneumatic system **310** and/or a working frequency of the pneumatic system **310**. The output pressure of the pneumatic system **310** determines a volume of the airbag **321** after expansion and a volume after deflation, and the working frequency of the pneumatic system **310** determines a period of the airbag **321** of switching between inflation and deflation to meet the massage needs of different users.

(52) In one embodiment, the vibrating assembly **200** is electrically connected with the control board **600** and the battery **700**. The battery **700** is configured to provide power to the pneumatic system **310** and the vibrating assembly **200**, and the control board **600** is configured to adjust a working condition of the pneumatic system **310** and a working condition of the vibrating assembly **200**. Specifically, the control board **600** accurately controls a working mode, the working frequency, and an intensity of the pneumatic system **310**. The control board **600** further accurately controls a working mode, a working frequency, and an intensity of the vibrating assembly **200**. Therefore, the user is able to adjust the massager to customize the massage effect. Specifically, the user is able to adjust parameters of the control board **600** to personalize the massage effect of the massager according to his/her need and preferences to obtain a more comfortable and satisfactory massage experience. In addition, the control board **600** may integrate a variety of massage modes and setting options, so that the massager has a variety of massage methods and functions. The user is able to choose different massage modes according to needs, such as choosing a vibration mode, choosing a pulse mode, adjusting a massage intensity, etc., to meet massage effects of different parts and health needs. Through intelligent control of the control board **600**, the massager provides the user a more personalized and diverse massage experience.

(53) As shown in FIGS. **13-14**, in one embodiment, the massager further comprises a button assembly **800**. The button assembly **800** is mounted on an outer surface of the mounting portion **110** and is electrically connected with the control board **600**. In some structural forms, a surface where the button assembly **800** is mounted on the mounting portion **110** is defined as the front surface of the massager body **100**, which is in line with the user's usage habits and enables the user to interact with the massager. The user is able to control the control panel **600** through the button assembly **800** to adjust settings and parameters of the massager, thereby enhancing the convenience and experience of the user. Of course, the user is able to adjust the settings and the parameters of the massager through a touch screen or a wireless remote control, which is not limited thereto.

(54) As shown in FIGS. **13-15**, in one embodiment, in order to facilitate the battery **700** to provide power to the vibrating assembly **200**, the massage portion **120** defines h a wiring channel **123**. The wiring channel **123** is communicated with the accommodating cavity **121** and the mounting cavity **113**. In this way, a first end of a wire assembly is extended into the accommodating cavity **121** through the wiring channel **123** and is electrically connected with the vibrating assembly **200**. A second end of the wire assembly is extended into the mounting cavity **113** and is electrically connected with the battery **700**. In this way, the vibrating assembly **200** is conveniently electrically connected with the battery **700**, and the wire assembly is mounted in the wiring channel **123** in an orderly manner without being messy. In one optional embodiment, the wiring channel **123** is defined in a middle portion of the massage portion **120**, so that the wire assembly is disposed in wiring channel and is stably protected by the wire assembly. In one alternative embodiment, the wiring channel **123** is a wiring groove defined in an outer wall of the massage portion **120** and an outer wall of the mounting portion **110**. Two opposite sides of the wiring groove are respectively communicated with the accommodating cavity **121** and the mounting cavity **113**. However, a specific form of the wiring channel **123** is not limited thereto.

(55) As shown in FIG. **15**, in one embodiment, the massage portion **120** comprises a third portion

124 and a fourth portion **125**. The third portion **124** is detachably connected with the fourth portion **125**. The third portion **124** defines the accommodating cavity **121**, and one end of the fourth portion **125** away from the first portion **412** is connected with the mounting portion **110**. The massage portion **120** comprises the third portion **124** and the fourth portion **125** that are detachably connected, so that the vibrating assembly **200** is conveniently mounted in the accommodating cavity **121**. In one embodiment, the second portion **413** defines the wiring channel **123** that is communicated with the accommodating cavity **121** and the mounting cavity **113**. In this way, the vibrating assembly **200** is electrically connected with the battery **700** through the wire assembly. In one alternative embodiment, since the third portion **124** and the fourth portion **125** are detachably connected, the vibrating assembly **200** is defined to have a power supply. In this way, the vibrating assembly **200** is not electrically connected with the battery **700** in the mounting cavity **113**, but vibrate through the power supply thereof. When the power supply of the vibrating assembly **200** is out of power, the third portion **124** and the fourth portion **125** are disassembled, and then the vibrating assembly **200** is taken out for charging. A specific form of the power supply of the vibrating assembly **200** is not limited thereto.
(56) The third portion **124** and the fourth portion **125** are detachably connected by snapping, screwing, etc., which is not limited thereto.
(57) As shown in FIG. **14**, in one embodiment, in order to facilitate mounting of above-mentioned components in the mounting cavity **113**, the mounting portion **110** of the present disclosure includes a first housing **111** and a second housing **112**. The first housing **111** is connected with the second housing **112**. The first housing **111** and the second housing **112** are enclosed to define the mounting cavity **113**. The first housing **111** and the second housing **112** are fixedly connected by one or more forms of screw connection, snap connection, and bonding. The first housing **111** and the second housing **112** are injection molded which is simple to process and enables first housing **111** and the second housing **112** to have complex shapes. The first housing **111** and the second housing **112** are made from, for example, but not limited to, polyethylene (PE), polypropylene (PP), polyvinyl chloride (PVC), polystyrene (PS), polycarbonate (PC), polyamide (PA), and polyester (PET).
(58) Specifically, the first housing **111** and/or the second housing **112** define vent holes communicated with the mounting cavity **113**. Thus, the air pump **311** is able to suck air from the outside through the mounting cavity **113** and the vent holes, while the control valve **312** and the pressure relief valve **315** release the air outward through the mounting cavity **113** and the vent holes, facilitating the inflation and deflation of the airbag **321**. The vent holes are define on one side of the first housing **111** away from the massage portion **120** and/or one side of the second housing **112** away from the massage portion **120**. That is, the vent holes are defined on a bottom portion of the massager body **100**. Moreover, an opening of the flexible sleeve **400** is designed to be corresponding to location of the vent holes.
(59) As shown in FIG. **13**, in one embodiment, a joint between the massage portion **120** and the mounting portion **110** is smoothly transitioned to reduce a risk of skin irritation or compression from edges or corners of the massager, thereby reducing a risk of skin injury. Especially when the massager vibrates at a high frequency, the joint that is smoothly transitioned effectively protects the skin from irritation or damage. Additionally, the joint that is smoothly transitioned further improves overall stability of the massager. By such design, a combination between the massage portion **120** and the mounting portion **110** is tighter, reducing a possibility of loosening or falling off, and making the massager stable and reliable when in use. Furthermore, the joint that is smoothly transitioned enhances aesthetic appeal of the massager. Whether in terms of appearance design or material texture, the joint that is smoothly transitioned enhance the aesthetics of the massager and make the massager more in line with aesthetic needs of the user.
(60) As shown in FIGS. **1-2**, in some embodiments, the massage portion **120** and the mounting portion **110** extend in substantially the same direction, so that the user is able to hold the mounting

portion **110**. As shown in FIGS. **16-19**, in some embodiments, the massage portion **120** and the mounting portion **110** extend in different directions. For instance, the mounting portion **110** is curved and arched, and the massage portion **120** is mounted on one side of the mounting portion **110** and is disposed in a mounting groove defined by the mounting portion **110**. In these embodiments, a length of the massager is reduced for conveniently transportation and packaging.
(61) In the drawings of the embodiments, the same or similar numbers correspond to the same or similar components. In the description of the present disclosure, it should be understood that terms such as “upper”, “lower”, “left”, “right”, etc., indicate direction or position relationships shown based on the drawings, and are only intended to facilitate the description of the present disclosure and the simplification of the description rather than to indicate or imply that the indicated device or element must have a specific direction or constructed and operated in a specific direction. Therefore, the terms used to describe positional relationships in the drawings are only for illustrative purposes and cannot be construed as limitations of the present disclosure. For those of ordinary skill in the art, the specific meanings of the above terms can be understood according to specific circumstances.
(62) The above are only optional embodiments of the present disclosure and are not intended to limit the present disclosure. Any modifications, equivalent substitutions, and improvements made within the spirit and principles of the present disclosure shall be included in the protection scope of the present disclosure.

Claims

1. A massager, comprising: a massager body; a vibrating assembly; and an expansion massage structure; wherein the massager body comprises a mounting portion and a massage portion connected with the mounting portion, the mounting portion defines a mounting cavity, the massage portion defines an accommodating cavity, and the massage portion is configured to insert into a human body cavity to massage the human body cavity; wherein the vibrating assembly is mounted in the accommodating cavity, located at a distal end of the accommodating cavity away from the mounting portion, and is configured to drive the massage portion to vibrate; wherein the expansion massage structure is disposed at a side of the vibrating assembly near the mounting portion and comprises a pneumatic system and at least one expansion portion, the pneumatic system is mounted in the mounting cavity and is configured to drive the at least one expansion portion to expand and contract, the at least one expansion portion is disposed on a periphery of the massage portion, and the at least one expansion portion is configured to insert into the human body cavity and massage the human body cavity; wherein the at least one expansion portion comprises an airbag, the massager further comprises a flexible sleeve, the flexible sleeve comprises a main portion, the main portion is assembled on an outer side of the massager body, and the airbag is formed on the flexible sleeve; wherein along a length direction of the flexible sleeve, two opposite ends of the airbag are connected with the main portion to define a gap between a middle portion of the airbag and the main portion.
2. The massager according to claim 1, wherein the massager body has a front surface, side surfaces, and a rear surface, and the at least one expansion portion is disposed on the front surface of the massager body; and/or the at least one expansion portion is disposed on any one of the side surfaces of the massager body; and/or the at least one expansion portion is disposed on the rear surface of the massager body.
3. The massager according to claim 2, wherein the at least one expansion portion comprises two expansion portions, the two expansion portions are respectively disposed on the side surfaces of the massager body; or the two expansion portions are respectively disposed on the front surface and the rear surface of the massager body.
4. The massager according to claim 1, wherein the pneumatic system comprises an air pump, a

control valve, and a connecting tube, an air outlet of the air pump is communicated with an air inlet of the control valve, an air outlet of the control valve is communicated with the connecting tube, and the airbag is connected and communicated with the connecting tube.
5. The massager according to claim 4, wherein the pneumatic system is communicated with the airbag and is not fluidly-communicated with the main portion.
6. The massager according to claim 5, wherein the airbag comprises a first end and a second end disposed opposite to the first end, the main portion comprises a first portion and a second portion connected with the first portion, the first portion is configured to sleeve on an outer side of the mounting portion, and the second portion is configured to sleeve on an outer side of the massage portion; wherein the main portion defines a first assembly cavity, and the first assembly cavity is configured to accommodate the massager body; wherein the first end of the airbag is connected with the first portion, and the airbag is communicated with the first assembly cavity through the first end thereof, the second end of the airbag is connected with the second portion, and a gap is formed between the airbag and the main portion.
7. The massager according to claim 4, wherein the pneumatic system further comprises a pressure relief valve, the pressure relief valve is communicated with the airbag and an outside, and when an air pressure in the airbag is greater than a predetermined value, the pressure relief valve is configured to reduce the air pressure in the airbag.
8. The massager according to claim 7, wherein the pressure relief valve comprises a valve housing, a reset piece, and a valve core; wherein the valve housing defines a valve cavity, a first opening, and a second opening, the first opening and the second opening are communicated with the valve cavity, the reset piece and the valve core are mounted in the valve cavity, the first opening is communicated with the airbag, and the second opening is communicated with the outside; wherein when driven by the reset piece, the valve core is configured to close the first opening; and when the air pressure in the airbag is greater than the predetermined value, the valve core is opened by air in the airbag.
9. The massager according to claim 7, wherein the pneumatic system further comprises a connecting piece, the connecting piece defines a first channel, a second channel, and a third channel, wherein the first channel, the second channel, and the third channel are communicated with each other, the first channel is communicated with the connecting tube, the second channel is communicated with the airbag, and the third channel is communicated with the pressure relief valve.
10. The massager according to claim 1, wherein the massage portion has a first surface facing towards the at least one expansion portion; the first surface is bent away from the at least one expansion portion; when the at least one expansion portion is not expanded, a gap is formed between the first surface and the at least one expansion portion; and when the at least one expansion portion is expanded, the at least one expansion portion occupies the gap.
11. The massager according to claim 1, wherein the massager further comprises a battery and a control board, the battery and the control board are mounted in the mounting cavity, and the control board is configured to adjust a working condition of the pneumatic system.
12. The massager according to claim 11, wherein the control board is configured to adjust an output pressure of the pneumatic system and/or a working frequency of the pneumatic system.
13. The massager according to claim 11, wherein the massager further comprises a button assembly, and the button assembly is mounted on an outer surface of the mounting portion and is electrically connected with the control board.
14. The massager according to claim 1, wherein the mounting portion comprises a first housing and a second housing, the first housing and the second housing are connected with each other, the first housing and the second housing are enclosed to form the mounting cavity, the first housing and/or the second housing define vent holes communicated with the mounting cavity.
15. A massager, comprising: a massager body; a vibrating assembly; and an expansion massage

structure; wherein the massager body comprises a mounting portion and a massage portion connected with the mounting portion, the mounting portion defines a mounting cavity, the massage portion defines an accommodating cavity, and the massage portion is configured to insert into a human body cavity to massage the human body cavity; wherein the vibrating assembly is mounted in the accommodating cavity, located at a distal end of the accommodating cavity away from the mounting portion, and is configured to drive the massage portion to vibrate; wherein the expansion massage structure is disposed at a side of the vibrating assembly near the mounting portion and comprises a pneumatic system and at least one expansion portion, the pneumatic system is mounted in the mounting cavity and is configured to drive the at least one expansion portion to expand and contract, the at least one expansion portion is disposed on a periphery of the massage portion, and the at least one expansion portion is configured to insert into the human body cavity and massage the human body cavity; wherein the at least one expansion portion comprises an airbag and a protection portion assembled on an outer side of the airbag; the massager further comprises a flexible sleeve, and the flexible sleeve comprises a main portion; wherein along a length direction of the flexible sleeve, two opposite ends of the protection portion are connected with the main portion to define a gap between a middle portion of the protection portion and the main portion.

16. A massager, comprising: a massager body; a vibrating assembly; and an expansion massage structure; wherein the massager body comprises a mounting portion and a massage portion connected with the mounting portion, the mounting portion defines a mounting cavity, the massage portion defines an accommodating cavity, and the massage portion is configured to insert into a human body cavity to massage the human body cavity; wherein the vibrating assembly is mounted in the accommodating cavity, located at a distal end of the accommodating cavity away from the mounting portion, and is configured to drive the massage portion to vibrate; wherein the expansion massage structure is disposed at a side of the vibrating assembly near the mounting portion and comprises a pneumatic system and at least one expansion portion, the pneumatic system is mounted in the mounting cavity and is configured to drive the at least one expansion portion to expand and contract, the at least one expansion portion is disposed on a periphery of the massage portion, and the at least one expansion portion is configured to insert into the human body cavity and massage the human body cavity; wherein the massage portion has a first surface facing towards the at least one expansion portion; the first surface is bent away from the at least one expansion portion; when the at least one expansion portion is not expanded, a gap is formed between the first surface and the at least one expansion portion; and when the at least one expansion portion is expanded, the at least one expansion portion occupies the gap.